

Supplementary Materials:
An End-to-End Multimodal System for Subtitle Recognition and
Chinese–Japanese Translation with Speech Synthesis in Short
Dramas

Jing An, Qiming Bao, and Jinhua Su

This document collects supporting tables, figures, and the fusion pseudocode referenced from the main manuscript. Numbering is independent of the main paper.

Algorithm S1 Confidence-Adaptive OCR/ASR Fusion

Require: Whisper segments $\mathcal{A}$, OCR detections $\mathcal{O}$, time tolerance δ , log-prob threshold θ_p , similarity threshold θ_s

Ensure: Fused subtitle sequence $\mathcal{S}$

```
1:  $\mathcal{S} \leftarrow \emptyset$ 
2: for each  $a_i = (t_i^s, t_i^e, w_i, p_i) \in \mathcal{A}$  do
3:    $\mathcal{C}_i \leftarrow \{o_j \in \mathcal{O} \mid \tau_j \in [t_i^s - \delta, t_i^e + \delta]\}$  ▷ candidate OCR detections
4:   if  $p_i \geq \theta_p$  then ▷ ASR is confident; trust Whisper
5:      $\hat{w}_i \leftarrow w_i$ 
6:   else
7:      $o^* \leftarrow \arg \max_{o_j \in \mathcal{C}_i} \text{sim}(o_j, w_i)$  ▷ best OCR match
8:     if  $\mathcal{C}_i \neq \emptyset$  and  $\text{sim}(o^*, w_i) \geq \theta_s$  then
9:        $\hat{w}_i \leftarrow o^*$  ▷ substitute OCR text, retain ASR timing
10:    else
11:       $\hat{w}_i \leftarrow w_i$  ▷ fallback: keep ASR
12:    end if
13:  end if
14:   $\mathcal{S} \leftarrow \mathcal{S} \cup \{(t_i^s, t_i^e, \hat{w}_i)\}$ 
15: end for
16: return  $\mathcal{S}$ 
```

Table S1: Statistical information of the proposed dataset.

Dataset Composition		Subtitle Statistics	
Video clips (total)	82	Total duration (min)	130.56
Paired episodes	79	Subtitle segments	3,692
Speaker IDs	12	Avg. segments/episode	46.7

Table S2: Gating-rule ablation under a single unified evaluation harness (identical fusion implementation and corpus-level segment-joined metrics; all 79 episodes). “Static” rows open the log-probability gate for every segment ($\theta_p=0$); the Adaptive row applies the selected operating point ($\theta_p=-0.2, \theta_s=80$). All rows are directly comparable. Absolute values differ from the production-pipeline fusion metrics in the main paper due to the corpus-level text-joining format. Best per column in bold.

Fusion Strategy	BLEU $\uparrow$	chrF++ $\uparrow$	CER $\downarrow$	CharAcc $\uparrow$	Comp. $\uparrow$
Never Replace (ASR only)	81.31	67.20	0.156	0.898	0.8066
Always Replace ($\theta_s=0$)	0.51	14.07	0.872	0.138	0.1030
Static ($\theta_s=40$)	81.35	68.32	0.161	0.901	0.8091
Static ($\theta_s=60$)	81.48	68.12	0.158	0.900	0.8094
Static ($\theta_s=80$)	81.61	67.77	0.156	0.899	0.8092
Adaptive ($\theta_p=-0.2, \theta_s=80$, ours)	81.60	67.73	0.156	0.899	0.8091

Table S3: Frame sampling rate ablation for subtitle recall on episode 11.mp4 using Qwen3-VL-4B.

Sampling Rate	Detected Characters	Recall vs. 5 fps
1 fps	411	27.7%
2 fps	659	44.3%
5 fps	1,486	100% (upper bound)

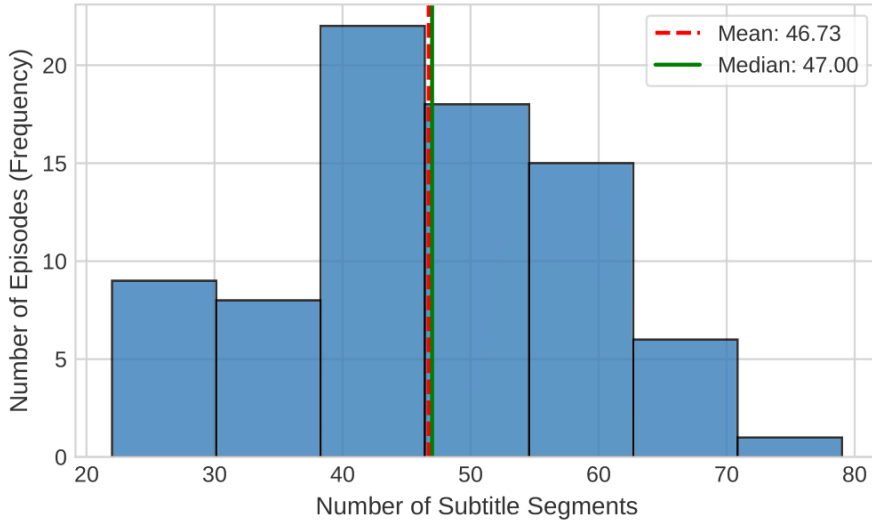


Figure S1: Distribution of subtitle segments per episode across the 79-episode dataset.

Table S4: Grid search composite scores on the 20-episode validation set. Each cell is the corpus-level composite score for the fusion strategy gated by (θ_p, θ_s) . The upper-left “flat” region ($\theta_p \leq -0.8$) corresponds to the ASR-always regime: only 1.6% of corpus segments have `avg_logprob` below -0.8 , so essentially no OCR substitutions occur. Scores are rounded to three decimal places; the bold marks the unique full-precision maximum ($\theta_p = -0.2$, $\theta_s = 80\%$: 0.8051 vs. 0.8046 at $\theta_s \in \{40\%, 70\%\}$, which also round to 0.805).

$\theta_p \setminus \theta_s$	40%	50%	60%	70%	80%
≤ -0.8	0.800	0.800	0.800	0.800	0.800
-0.5	0.802	0.801	0.800	0.800	0.800
-0.4	0.802	0.801	0.800	0.800	0.800
-0.3	0.804	0.803	0.802	0.802	0.803
-0.2	0.805	0.803	0.803	0.805	0.805

Table S5: Word-level error type analysis of adaptive fusion corrections (jieba POS; nr/ns/nt/nz = proper nouns). GT correction rate: fraction of substitutions that moved the output closer to the ground truth.

Category	Substitutions	% of Total	GT Corrections	Correction Rate
Proper nouns (nr/ns/nt/nz)	30	13.2%	18	60.0%
Numbers	4	1.8%	3	75.0%
Common vocabulary	193	85.0%	114	59.1%
Total	227			

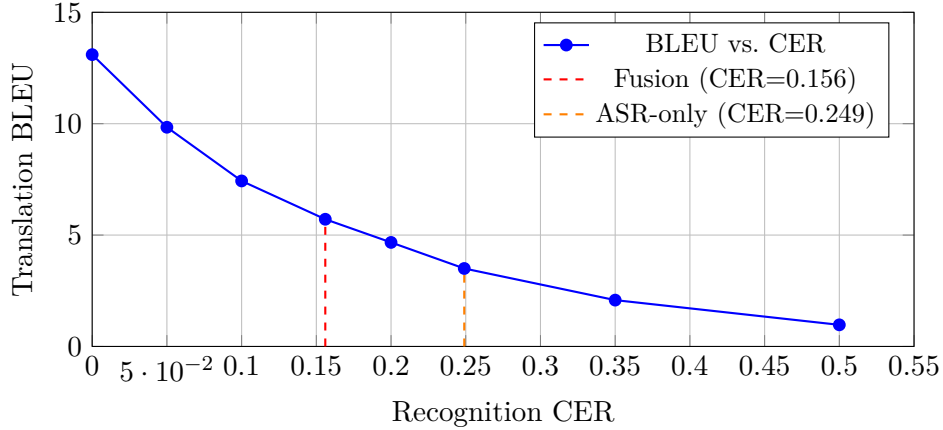


Figure S2: Cross-stage error propagation: translation BLEU as a function of recognition CER. Dashed lines mark the operating points of ASR-only (CER=0.249) and Adaptive Fusion (CER=0.156). The fusion’s recognition improvement yields +2.2 BLEU in the translation stage.

Table S6: End-to-end pipeline evaluation on 16 fold-0 validation episodes. Source text fed to the Qwen3-VL-4B LoRA translation model varies by condition; metrics are computed at the document level (entire episode as one unit) against the Japanese ground truth. chrF++ is the primary metric (character-level, more robust than BLEU for Japanese at document granularity). The ASR-only BLEU of 0.00 is a genuine consequence of propagated proper-noun errors eliminating all document-level 4-gram matches, not an evaluation artefact (see main text).

Source Condition	BLEU↑	chrF++↑
Oracle (gold Chinese text)	5.02	14.97
Adaptive Fusion (ours)	0.48	13.56
ASR-only (Whisper-medium)	0.00	11.32

Table S7: Translation quality ablation: visual context vs. text-only input using Qwen3-VL zero-shot on 20 episodes. Absolute BLEU gain: +3.35.

Input Modality	BLEU↑	chrF++↑
Text-Only (OCR subtitles)	10.09	28.26
Multimodal (Video frames + Text)	13.44	29.70

Table S8: TTS reference audio duration ablation using F5-TTS. Speaker similarity measured via cosine similarity of speaker embeddings.

Reference Duration	Speaker Similarity↑
Short (~3s)	0.64
Long (~10s)	0.71

Table S9: Qualitative translation examples. Example A shows a culturally nuanced term translated correctly by FT but incorrectly by ZS. Example B shows an OOV-driven hallucination by FT that ZS avoids.

Ex.	Chinese source	FT (ours)	ZS (Qwen3-VL-4B)	Reference
A	看来我们挺有缘	きっと縁があるな (cultural term preserved)	きっと運が良いのだろう (semantic error: 縁 → 運)	やっぱり縁があるのかもね
B	拜托	お前はお母さんも有名なファッションデザイナーの White さんだろ (OOV hallucination)	お祈り	っていうか